

Extrauterine Pregnancy 01.

Location Is the Diagnosis

Companion reading handout for simultaneous listening.

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Transcript

Extrauterine Pregnancy audio course. Episode 1. Location Is the Diagnosis.

A patient is seven weeks pregnant. The ultrasound report says, "gestational sac seen in the uterus." That phrase can be reassuring, but it is not yet the diagnosis you need. If the sac is centered in the endometrial cavity, surrounded by the normal implantation environment, that is one situation. If the sac is low and anterior, embedded in a cesarean scar niche with only a thin layer of myometrium toward the bladder, that is another situation entirely. Both can be described loosely as intrauterine. Only one is normally sited.

Stay with that for a moment, because this is the whole course in miniature. The uterus is an organ, but safe early implantation is not defined by the outer border of the organ. It is defined by the endometrial cavity, the endometrial-myometrial junction, and the decidual tissue that normally regulates trophoblast. A pregnancy can be inside the uterine outline and still be in the wrong tissue plane. That is why the word "intrauterine" can be too blunt for modern ectopic pregnancy care.

That distinction is the starting point for modern ectopic pregnancy care.

An ectopic pregnancy is not best understood as simply "a pregnancy outside the uterus." That older shorthand works for most cases because most ectopic pregnancies are tubal. ACOG states that the fallopian tube accounts for more than 90 percent of ectopic implantations, and Nature Reviews summarizes that tubal pregnancies form the large majority of ectopic pregnancies. But the shorthand fails exactly where the stakes can be highest. A pregnancy may be anatomically within the uterus and still be outside the correct physiological implantation site.

The better definition is anatomical and functional: ectopic pregnancy is implantation outside the uterine endometrial cavity, or outside the normally sited endometrial implantation environment. Farren and colleagues make this point sharply in their modern review: the important boundary is not the outer wall of the uterus, but the normal implantation site defined by the endometrial cavity and the endometrial-myometrial junction. ESHRE similarly pushes terminology toward pregnancy location rather than vague reassurance.

The reason is practical. The implantation bed determines what the pregnancy can do to the patient.

The endometrium and decidua are not passive lining. They are the maternal tissue system built to receive trophoblast, regulate invasion, support vascular remodeling, and later separate in a controlled way. The fallopian tube does not have that capacity. The cervix does not clamp down like the uterine corpus. A cesarean scar niche may have deficient myometrium, altered vascularity, and a tissue plane that can behave like early abnormal placentation. The peritoneal cavity, ovary, rudimentary horn, and myometrial wall each create a different hemorrhage and rupture problem.

That is why the same embryo, at the same gestational age, with the same hCG, can mean different clinical danger in different locations. In the endometrial cavity, trophoblast is meeting tissue built for invasion. In the tube, trophoblast is meeting a transport organ. In the cervix, it is meeting tissue that bleeds but does not contract well. In a scar niche, it is meeting damaged myometrium near bladder and major vascular supply. The pathology is not only the pregnancy. The pathology is the conversation between trophoblast and the tissue it has invaded.

So location is not a label placed after diagnosis. Location is part of the diagnosis.

Start with normally sited pregnancy. This means the gestation is implanted in the uterine endometrial cavity. In spoken clinical language, "intrauterine pregnancy" often tries to mean this, but the term is no longer precise enough by itself. A cesarean scar pregnancy is also within the uterus. A cervical pregnancy is within the uterus. An intramural pregnancy is within the uterus. The word "intrauterine" can therefore hide the very abnormality that needs to be recognized.

Next, separate ectopic pregnancies into extrauterine and uterine ectopic pregnancies.

Extrauterine ectopic pregnancy includes the familiar tubal sites: ampullary, isthmic, and interstitial. It also includes ovarian and abdominal pregnancy. The ampulla is the commonest tubal site because it is the wider distal segment where fertilization commonly occurs and where delayed embryo transport can leave the conceptus long enough for implantation. Isthmic and interstitial pregnancies are less common, but the closer the pregnancy is to the uterus, the more surrounding muscle and blood supply can change the timing and severity of rupture.

Uterine ectopic pregnancy means the pregnancy is within the uterus but not normally sited in the endometrial cavity. This includes cesarean scar pregnancy, cervical pregnancy, and intramural pregnancy. These are not minor variants of tubal ectopic pregnancy. Their treatment and risk depend on the tissue they invade. Cesarean scar pregnancy is increasingly treated as an early abnormal placentation problem because the trophoblast is implanted in scar tissue near the lower uterine segment and bladder. Cervical pregnancy is dangerous because removal can cause severe hemorrhage from tissue that is poor at contracting for hemostasis. Intramural

pregnancy is implanted in the myometrium itself and can be difficult to diagnose because it may look close to the cavity while not being inside it.

Now add the temporary category: pregnancy of unknown location.

Pregnancy of unknown location, or PUL, is not a diagnosis. It is a statement about what is known at a moment in time. The pregnancy test is positive, the patient is clinically stable enough for follow-up, and transvaginal ultrasound has not yet demonstrated either a normally sited pregnancy or an ectopic pregnancy. PUL should never be used to make uncertainty sound finished. It is a holding category that requires a plan.

This deserves repetition because it is easy to lose in practice. PUL does not mean "probably fine." It does not mean "too early, go home" unless the patient is stable, has clear follow-up, and understands return precautions. PUL means that the pregnancy has not yet declared its address. The clinician's duty is to keep the patient safe while time, hCG, and repeat imaging allow the address to become clear.

The word stable is essential. A patient with a positive pregnancy test, severe abdominal pain, syncope, peritoneal signs, hypotension, or concerning hemoperitoneum is not merely a neat PUL. She is a possible ruptured ectopic pregnancy until proven otherwise. In a stable patient, uncertainty can be managed with time, repeat hCG, repeat ultrasound, and clear return precautions. In an unstable patient, the physiology becomes the diagnosis that matters first.

This terminology prevents three common errors.

The first error is false reassurance from a uterine address. If the report says there is a gestational sac in the uterus, the next question is whether it is normally sited within the endometrial cavity. A low anterior sac in a previous cesarean scar is not the same finding as a normally implanted gestational sac.

The second error is false reassurance after assisted reproduction. Heterotopic pregnancy means a normally sited pregnancy and an ectopic pregnancy coexist. A normally sited pregnancy can produce the hCG and can satisfy the eye on ultrasound while a tubal ectopic is still present. This is rare in spontaneous conception but clinically important after IVF or ovulation induction. In that setting, finding one pregnancy does not always finish the scan.

The third error is treating all ectopic pregnancies as if they share the same natural history. A small failing ampullary ectopic with falling hCG may be appropriate for observation in a reliable patient. A live interstitial pregnancy carries a different rupture and hemorrhage risk. A cesarean scar pregnancy may progress toward placenta accreta spectrum-level morbidity if continued. A cervical pregnancy can bleed catastrophically if mistaken for miscarriage tissue and curetted without planning.

At the bedside, the first task is to force the wording to become anatomical.

Do not ask, "Is there something in the uterus?" Ask, "Is the pregnancy normally sited in the endometrial cavity?" If not, ask exactly where the trophoblast is implanted. Tube, ovary, abdomen, cervix, scar, myometrium, rudimentary horn, or not yet localized.

Then ask what that site can safely tolerate. Can it distend? Can it regulate trophoblastic invasion? Can it contract if the pregnancy separates? Is it near large vessels? Is it near the bladder? Is the patient stable enough for time to clarify the diagnosis?

This is also how the resident should read an ultrasound report. A good report should not only say whether the uterus is empty. It should describe the location of the gestational sac or trophoblast, its relationship to the endometrial cavity, cervix, scar, myometrium, ovaries, and adnexa, and whether there is free fluid suggesting bleeding. If those details are missing and the clinical question is serious, the answer is not to fill the gaps with optimism. The answer is expert review, repeat imaging, or escalation according to the patient's condition.

The course will later discuss hCG, ultrasound signs, methotrexate, surgery, cesarean scar pregnancy, special-site ectopics, emergencies, and follow-up. But all of those decisions begin here. hCG tells you about trophoblast over time. Ultrasound tries to show anatomy and bleeding. Treatment depends on stability, tissue bed, active trophoblast, fertility goals, and follow-up reliability. None of those can be interpreted well if the location language is vague.

So when you listen to later thresholds and regimens, keep translating them back to location. Methotrexate treats active trophoblast, but whether methotrexate is safe depends on whether the tissue bed is quiet enough to watch. Surgery removes or controls the implantation site, but the operation depends on what the site is. Follow-up follows hCG, but the danger during follow-up still comes from the tissue bed. The language of location is the scaffold that holds all later details.

So the first discipline is linguistic because the first safety act is anatomical precision.

When you hear "intrauterine," do not relax until you know whether the pregnancy is normally sited. When you hear "PUL," do not treat it as a diagnosis; treat it as a follow-up obligation. When you hear "ectopic," do not assume tubal; ask which implantation bed is carrying the risk.

Inside the uterus is not the same as normally implanted. Unknown location is not the same as low risk. And ectopic pregnancy is not one disease behavior. It is abnormal implantation, and the site of implantation writes the danger.